

Lecture 4

Fisher randomization inference: The California ballot-placement case study

How does the assignment mechanism determine the analysis?

Connection to R Example 1

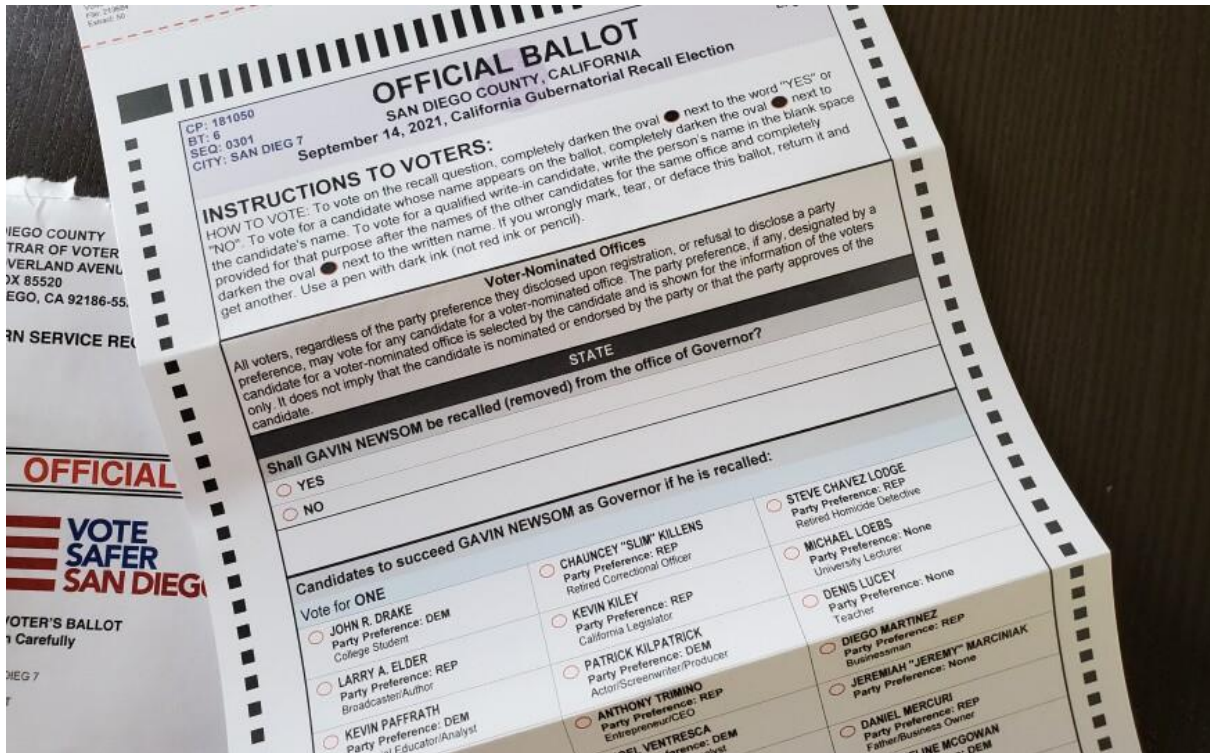
The same testing steps apply

1. State a sharp null and impute missing potential outcomes.
2. Generate assignments using the study's design.
3. Recompute the chosen statistic for each assignment.
4. Compare the observed result with this reference distribution.

In R Example 1, `sample(W)` preserves the treated count.
What should replace that step in the ballot study?

Does first-page placement increase vote share?

The study compares first-page placement with later-page placement for a particular candidate in the 2003 recall election.



135 candidates

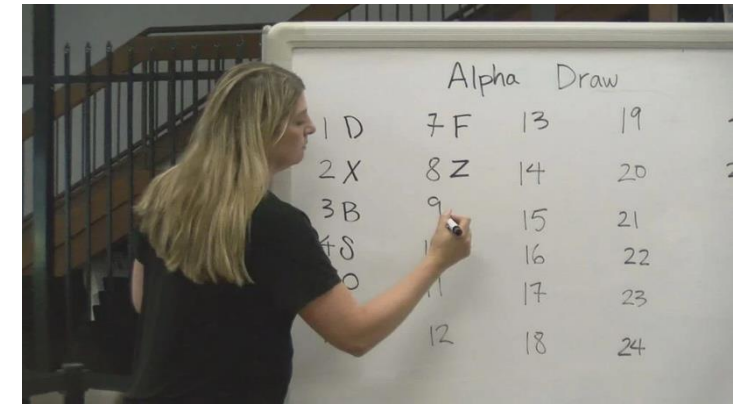
Could visibility matter more for unfamiliar candidates?

What could our units be: candidates, voters, or geographic areas?

The alphabet lottery and district rotation

California has 80 State Assembly districts, numbered 1–80.

1. Draw a randomized alphabet.
2. Use it to order names in District 1.
3. For District 2, move the first name to the end.
Repeat this rotation through District 80.



County ballot formats then determine each candidate's page.

Once the alphabet is drawn, which remaining steps are random?

A small rotation example

Illustration: four candidates, three districts, two names per page.
A lottery gives the initial order Chen, Adams, Diaz, Brown.

District	Candidate order	First page
1	Chen, Adams, Diaz, Brown	Chen, Adams
2	Adams, Diaz, Brown, Chen	Adams, Diaz
3	Diaz, Brown, Chen, Adams	Diaz, Brown

In which districts does Chen appear on the first page?

Each district uses the same two-page format in this illustration.

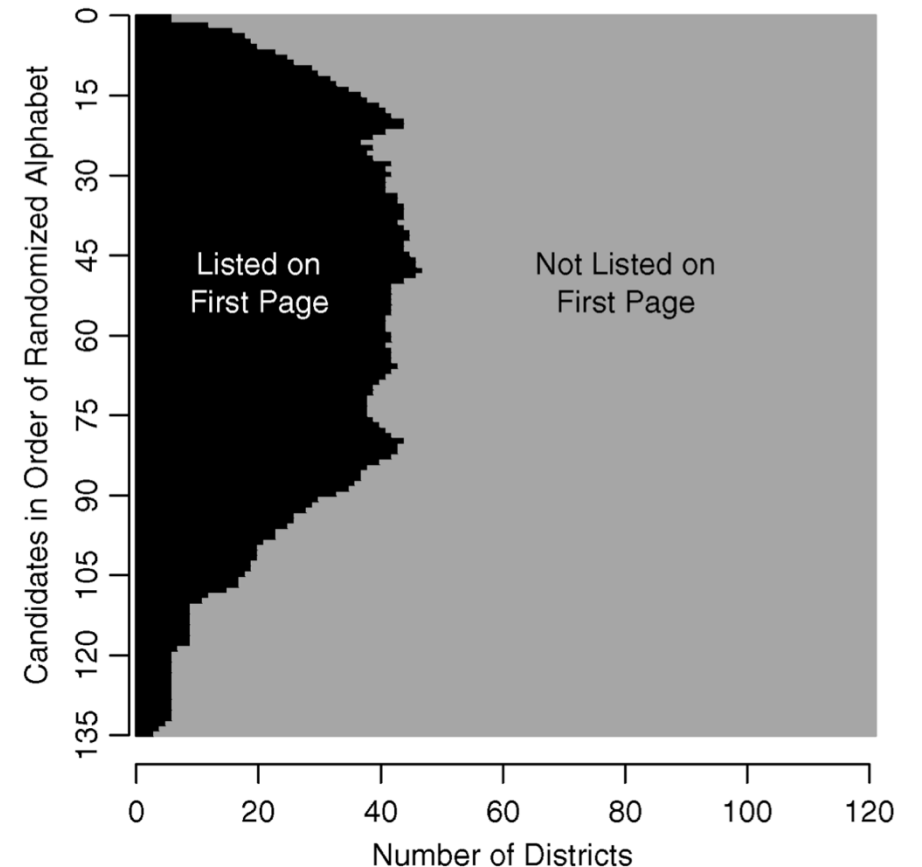
First-page exposure varies across candidates

135 candidates
80 assembly districts
58 county ballot formats

121 county–district combinations
have multipage ballots.

Page placement depends on both
district order and county format.

Should we compare different candidates in the same area,
or the same candidate across areas?



Compare across candidates or across districts?

Option 1: different candidates within one county–district

Units: candidates. Outcome: each candidate's vote share.

If one candidate gains votes from placement, could another lose votes?

Is no interference between these units plausible?

Option 2: one candidate across county–districts

Units: geographic areas. Outcome: that candidate's local vote share.

Could placement in one area affect their vote share elsewhere?

Which comparison makes no interference more plausible? Why?

One candidate, compared across districts

The authors analyze each candidate separately.

Unit i : a county–district combination with a multipage ballot.

Outcome: that candidate's vote share in unit i .

Treatment W_i : 1 for first page, 0 for a later page.

$Y_i(1)$: vote share with first-page placement

$Y_i(0)$: vote share with later-page placement

Why is a county–district combination needed, rather than an assembly district alone?

Chen's assignment vector

Now take the three districts in our example as units.
 $W_i = 1$ means Chen is on the first page in district i .

Initial order: Chen, Adams, Diaz, Brown
Chen's first-page indicators: $W = (1, 0, 0)$

$\text{sample}(W)$ shuffles these indicators across districts.
It always keeps exactly one first-page placement.

Suppose Chen starts second: Adams, Chen, Diaz, Brown.
What is the new W ? Can $\text{sample}(c(1, 0, 0))$ generate it?

Assumptions behind the comparison

Known joint assignment mechanism

Reproduce the lottery, rotation, and fixed page formats.

Page formats must not be redesigned in response to the drawn alphabet.

No interference across units

A candidate's vote share in one district does not depend on that candidate's placement in another district.

Treatment versions

Does “first page versus later page” adequately describe exposure if exact position and neighboring candidates also matter?

Fisher's test follows the lottery

Sharp null for one candidate: $Y_i(1) = Y_i(0)$ for every unit i .
Under this null, keep the candidate's observed vote shares fixed.

Statistic D = mean vote share on first page – mean on later pages

For each simulation:

1. Draw an alphabet and sort the candidate names.
2. Rotate the order and apply the fixed county page formats.
3. Obtain the candidate's new W and recalculate D .

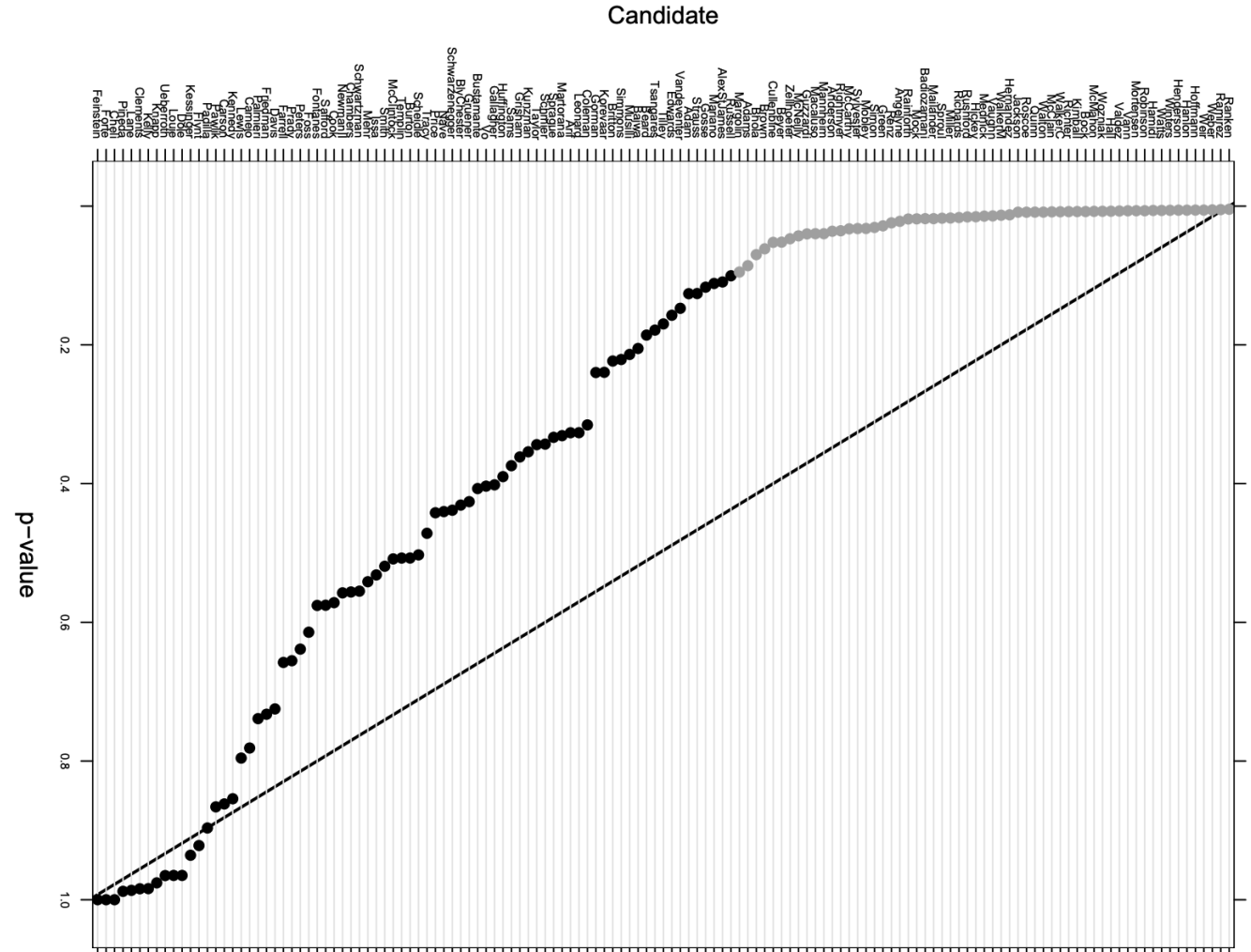
For evidence of an increase, count simulated $D \geq D$ observed.
The design supplies the reference distribution.

Evidence against the sharp null

One-sided p-values,
approximated by Monte Carlo.

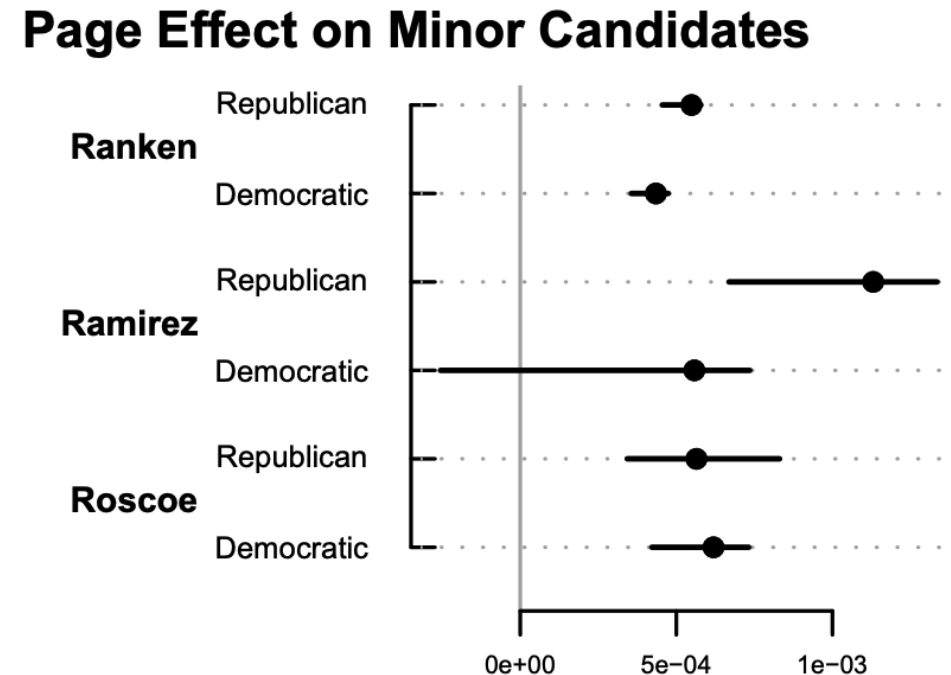
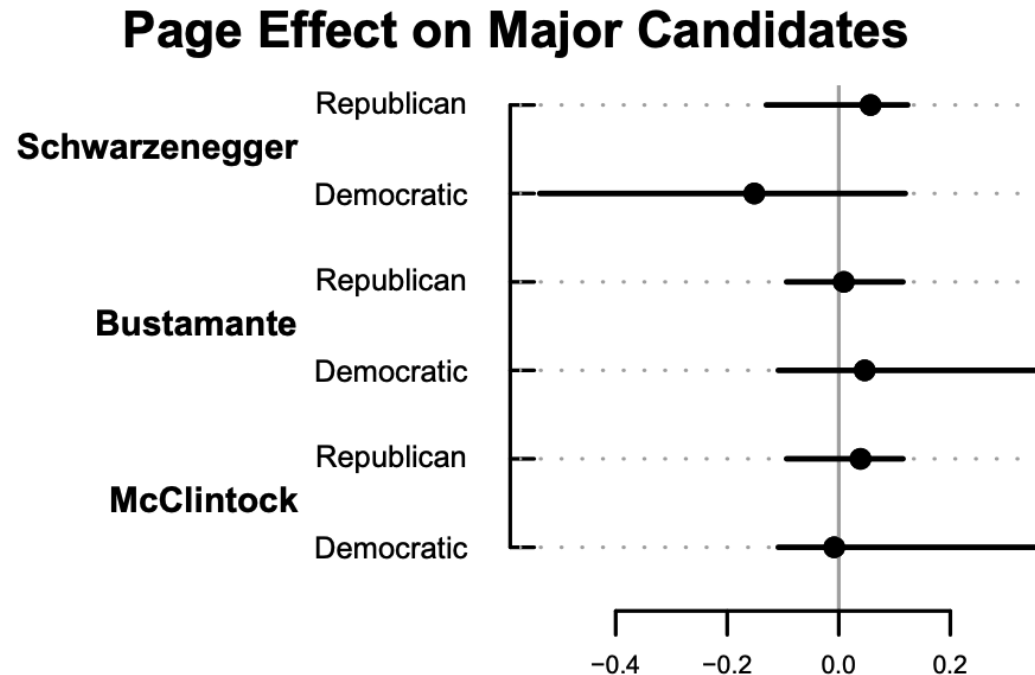
Smaller p-values appear
near the TOP of this figure.

59 of 135 tests reject at p-value
cutoff 0.01.



Effect sizes under a common-effect model

- For each candidate and district subgroup, assume $Y_i(1) - Y_i(0) = \tau$ across all republican / democratic districts.
- Impute under each candidate τ_0 , repeat the lottery, and invert the tests.



90% intervals; effects are differences in vote shares. Panel scales differ.

What carries over to other studies?

The sharp null supplies the missing outcomes.

The actual assignment mechanism supplies the reference distribution.

A common-effect confidence interval requires an additional assumption.

What would go wrong if we replaced the alphabet lottery with $\text{sample}(W)$ across the observed districts?